

HW 1

Erin Cooper

2026-09-09

#The following code is used to open the lead dataset for the 9-9-2026 project, and to load it to a csv.

```
lead <- read.csv("C:\\Users\\erinc\\OneDrive - Colorado School of Mines\\Desktop\\Statistical Consulting\\lead.csv")
#View(lead)
```

#replaces the outlier value with 99 instead of 999

```
outlier <- which(lead$IQ == 999)
```

```
lead[outlier,2] = 99
```

#the following code intends to create a graph showing the IQ levels by location status

```
ggplot(data = lead, aes(x = Smelter, y = IQ))+geom_boxplot()+labs(title = "Smelter distance vs. IQ")
```

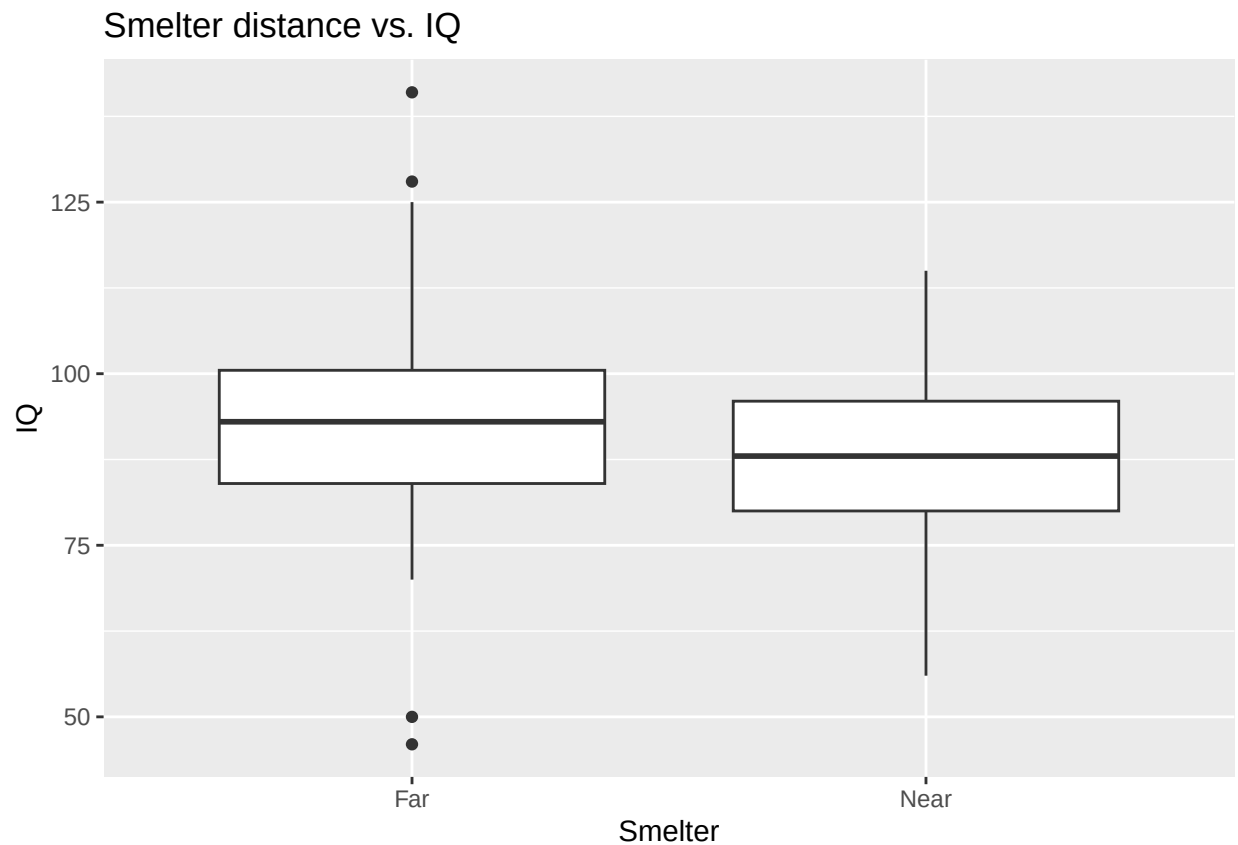


Table 1: A table presenting summary statistics for participants in the study.

Smelter	Median	Mean	Number of Participants
Far	93	92.69	67
Near	88	89.19	57

```
#the following code intends to create a nice looking table
table = lead %>% drop_na(Smelter, IQ) %>% group_by(Smelter) %>%
  summarise(
    Median = median(IQ),
    Mean = round(mean(IQ), 2),
    n = n())

kbl(table, col.names = c("Smelter", "Median", "Mean", "Number of Participants"), booktabs = TRUE, capti
```

The graph above displays the distribution of the IQ's of the participants depending on their proximity to the smelter. The mean of the overall IQ was: 91.0806452.

The table above displays the smelter distance, as well as the median and mean of the participants. The median was added in addition to the mean because one of the data points is significantly different from the others which could skew the mean.

AI use disclaimer

The Google AI search function was used to remember the function for the box plot function in ggplot.